

A RESULTS ON EVALUATION OF GENERATIVE ADVERSARIAL NETWORKS

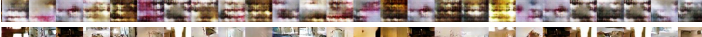
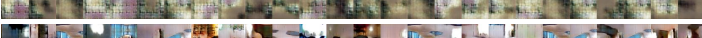
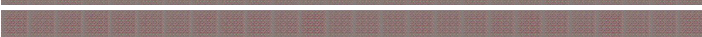
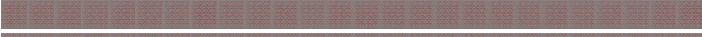
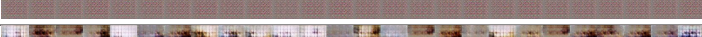
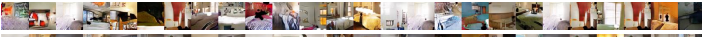
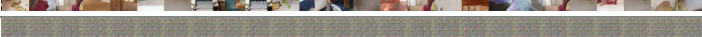
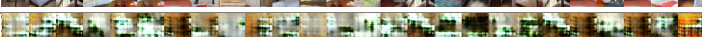
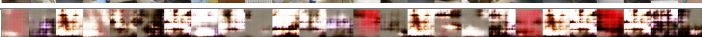
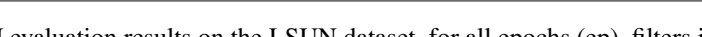
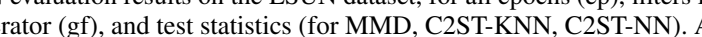
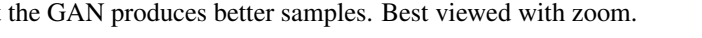
gf	df	ep	random sample	MMD	KNN	NN
-	-	-		-	-	-
32	32	1		0.154	0.994	1.000
32	32	10		0.024	0.831	0.996
32	32	50		0.026	0.758	0.983
32	32	100		0.014	0.797	0.974
32	32	200		0.012	0.798	0.964
32	64	1		0.330	0.984	1.000
32	64	10		0.035	0.897	0.997
32	64	50		0.020	0.804	0.989
32	64	100		0.032	0.936	0.998
32	64	200		0.048	0.962	1.000
32	96	1		0.915	0.997	1.000
32	96	10		0.927	0.991	1.000
32	96	50		0.924	0.991	1.000
32	96	100		0.928	0.991	1.000
32	96	200		0.928	0.991	1.000
64	32	1		0.389	0.987	1.000
64	32	10		0.023	0.842	0.979
64	32	50		0.018	0.788	0.977
64	32	100		0.017	0.753	0.959
64	32	200		0.018	0.736	0.963
64	64	1		0.313	0.964	1.000
64	64	10		0.021	0.825	0.988
64	64	50		0.014	0.864	0.978
64	64	100		0.019	0.685	0.978
64	64	200		0.021	0.775	0.980
64	96	1		0.891	0.996	1.000
64	96	10		0.158	0.830	0.999
64	96	50		0.015	0.801	0.980
64	96	100		0.016	0.866	0.976
64	96	200		0.020	0.755	0.983
96	32	1		0.356	0.986	1.000
96	32	10		0.022	0.770	0.991
96	32	50		0.024	0.748	0.949
96	32	100		0.022	0.745	0.965
96	32	200		0.024	0.689	0.981
96	64	1		0.287	0.978	1.000
96	64	10		0.012	0.825	0.966
96	64	50		0.017	0.812	0.962
96	64	100		0.019	0.670	0.983
96	64	200		0.020	0.711	0.972
96	96	1		0.672	0.999	1.000
96	96	10		0.671	0.999	1.000
96	96	50		0.829	0.999	1.000
96	96	100		0.668	0.999	1.000
96	96	200		0.849	0.999	1.000

Table 5: GAN evaluation results on the LSUN dataset, for all epochs (ep), filters in discriminator (df), filters in generator (gf), and test statistics (for MMD, C2ST-KNN, C2ST-NN). A lower test statistic estimates that the GAN produces better samples. Best viewed with zoom.

gf	df	ep	random sample	MMD	KNN	NN
-	-	-		-	-	-
32	32	1		0.806	1.000	1.000
32	32	10		0.152	0.940	1.000
32	32	50		0.042	0.788	0.993
32	32	100		0.029	0.808	0.982
32	32	200		0.022	0.776	0.970
32	64	1		0.994	1.000	1.000
32	64	10		0.989	1.000	1.000
32	64	50		0.050	0.808	0.985
32	64	100		0.036	0.766	0.972
32	64	200		0.015	0.817	0.987
32	96	1		0.995	1.000	1.000
32	96	10		0.992	1.000	1.000
32	96	50		0.995	1.000	1.000
32	96	100		0.053	0.778	0.987
64	96	200		0.037	0.779	0.995
64	32	1		1.041	1.000	1.000
64	32	10		0.086	0.971	1.000
64	32	50		0.043	0.756	0.988
64	32	100		0.018	0.746	0.973
64	32	200		0.025	0.757	0.972
64	64	1		0.836	1.000	1.000
64	64	10		0.103	0.910	0.998
64	64	50		0.018	0.712	0.973
64	64	100		0.020	0.784	0.950
64	64	200		0.022	0.719	0.974
64	96	1		1.003	1.000	1.000
64	96	10		1.015	1.000	1.000
64	96	50		1.002	1.000	1.000
64	96	100		1.063	1.000	1.000
64	96	200		1.061	1.000	1.000
96	32	1		1.022	1.000	1.000
96	32	10		0.222	0.978	1.000
96	32	50		0.026	0.734	0.965
96	32	100		0.016	0.735	0.964
96	32	200		0.021	0.780	0.973
96	64	1		0.715	1.000	1.000
96	64	10		0.042	0.904	0.999
96	64	50		0.024	0.697	0.971
96	64	100		0.028	0.744	0.983
96	64	200		0.020	0.697	0.976
96	96	1		0.969	1.000	1.000
96	96	10		0.920	1.000	1.000
96	96	50		0.926	1.000	1.000
96	96	100		0.920	1.000	1.000
96	96	200		0.923	1.000	1.000

Table 6: GAN evaluation results on the LFW dataset, for all epochs (ep), filters in discriminator (df), filters in generator (gf), and test statistics (for MMD, C2ST-KNN, C2ST-NN). A lower test statistic estimates that the GAN produces better samples. Best viewed with zoom.

B PROOF OF THEOREM 1

Our statistic is a random variable $T \sim \mathcal{N}\left(\frac{1}{2}, \frac{1}{4n_{\text{te}}}\right)$ under the null hypothesis, and $T \sim \mathcal{N}\left(\frac{1}{2} + \epsilon, n_{\text{te}}^{-1}\left(\frac{1}{4} - \epsilon^2\right)\right)$ under the alternative hypothesis. Furthermore, at a significance level α , the threshold of our statistic is $z_\alpha = \frac{1}{2} + \frac{\Phi^{-1}(1-\alpha)}{\sqrt{4n_{\text{te}}}}$; under this threshold we would accept the null hypothesis. Then, the probability of making a type-II error is

$$\begin{aligned} \mathbb{P}_{T \sim \mathcal{N}\left(\frac{1}{2} + \epsilon, \frac{1}{4n_{\text{te}}}\right)}(T < z_\alpha) &= \mathbb{P}_{T' \sim \mathcal{N}\left(0, \frac{1}{4n_{\text{te}}}\right)}\left(T' < \frac{\Phi^{-1}(1-\alpha)}{\sqrt{4n_{\text{te}}}} - \epsilon\right) \\ &= \Phi\left(\sqrt{\frac{n_{\text{te}}}{\frac{1}{4} - \epsilon^2}}\left(\frac{\Phi^{-1}(1-\alpha)}{\sqrt{4n_{\text{te}}}} - \epsilon\right)\right) \\ &= \Phi\left(\frac{\Phi^{-1}(1-\alpha)/2 - \epsilon\sqrt{n_{\text{te}}}}{\sqrt{\frac{1}{4} - \epsilon^2}}\right). \end{aligned}$$

Therefore, the power of the test is

$$\pi(\alpha, n_{\text{te}}, \epsilon) = 1 - \Phi\left(\frac{\Phi^{-1}(1-\alpha)/2 - \epsilon\sqrt{n_{\text{te}}}}{\sqrt{\frac{1}{4} - \epsilon^2}}\right) = \Phi\left(\frac{\epsilon\sqrt{n_{\text{te}}} - \Phi^{-1}(1-\alpha)/2}{\sqrt{\frac{1}{4} - \epsilon^2}}\right),$$

which concludes the proof.

C ACKNOWLEDGEMENTS

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